∫ntegration

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*소개 및 진행방식 설명

- Integration Bee : 1981년, MIT를 시작으로 하여 여러 학교에서 열리고 있는 학부생들을 위한 적분 토너먼트 대회.
- 영어-식 단어 받아쓰기 대회인 Spelling Bee의 이름을 땄다.
- 문제는 10번까지 있으며 난이도에 따라 제한시간이 다르다.
- 가장 빠른 해답자들 몇 명과 제한시간 내 해답자들은 점수 획득.
- 총점 집계 후 1등에게는 수만 원 상당의 원하는 선물 지급(부회장이)

*규칙

- 문제가 띄워지는 순간부터 종이와 펜을 들고 풀기 시작.
- 적분값을 Zoom 채팅 중 DM 기능을 활용하여 ‘주하늘’로 전송
- 정오 여부는 제한시간 동안 공개되지 않음. 참가자는 제한시간 종료 시까지 자유롭게 답안 변경 가능
- 자신의 마지막 답안이 최종 답안. 정오 여부 및 제출 시각 체크
- 라운드 종료 시 해당 라운드의 결과 및 해설(crucial ctep) 공개

*안내사항

- 빠르고 정확한 답안 확인을 위하여, 회의실에서 본인의 이름을 공지해드리는 대로 A, B, C 등의 알파벳 대문자로 변경해주세요.

- 부정행위를 직접 확인할 방법은 없습니다.

  WolframAlpha 등의 계산 툴의 사용을 삼가해 주십시오.

  Note : $\int \frac{1}{1+x^2} dx = \tan^{-1}(x) + C$ 를 활용해 주세요..!!!

Problem 1

01:00

1 pt - All right answerers

$$\int_{23}^{29} 31x\,dx$$

$$\int_{23}^{29} 31x\,dx$$

$$\int_{23}^{29} 31x\,dx$$

$$\int_{23}^{29} 31x\,dx$$

$$\int_{23}^{29} 31x\,dx$$

$Time's\ up!$

- Answer : 4836
- Crucial Step :

  Knowing how to do an integral

  Submitting the answer properly

A	B	C	D	E	F	G	H	I	J	K	L

Problem 1

01:30

3 pts – Fastest 3, 1 pt – right answerers

$$\frac{2}{\pi}\int_0^{2021\pi} \sin^2 x\, dx$$

$$\frac{2}{\pi}\int_0^{2021\pi} \sin^2 x\, dx$$

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$$\frac{2}{\pi}\int_{0}^{2021\pi} \sin^2 x \, dx$$

$$\frac{2}{\pi}\int_{0}^{2021\pi} \sin^2 x \, dx$$

$Time's\ up!$

- Answer : 2021
- Crucial Step :

Getting your integrating brain ready

A	B	C	D	E	F	G	H	I	J	K	L

Problem 2

03:00

5 pts – Fastest 3, 3 pts – right answerers

$$8cos\left[\int_0^{\pi^2} sin(3\sqrt{x})\,dx\right]$$

$$8cos\left[\int_0^{\pi^2} sin(3\sqrt{x})\,dx\right]$$

$$8cos\left[\int_0^{\pi^2} sin(3\sqrt{x})\,dx\right]$$

$$8cos\left[\int_0^{\pi^2} sin(3\sqrt{x})\, dx\right]$$

$$8cos\left[\int_0^{\pi^2} sin(3\sqrt{x})\, dx\right]$$

$Time's\ up!$

- Answer : -4
- Crucial Step :

Substituting $u = \sqrt{x}$

Integrating by parts

A	B	C	D	E	F	G	H	I	J	K	L

Problem 3

04:00

5 pts – Fastest 3, 4 pts – right answerers

$$\pi \times \left[\int_0^{\frac{1}{3}} \frac{x}{1+x^4}\,dx + \int_1^3 \frac{x}{1+x^4}\,dx \right]^{-1}$$

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$$\pi \times \left[\int_0^{\frac{1}{3}} \frac{x}{1+x^4}\,dx + \int_1^3 \frac{x}{1+x^4}\,dx \right]^{-1}$$

$$\pi \times \left[\int_0^{\frac{1}{3}} \frac{x}{1+x^4}\, dx + \int_1^3 \frac{x}{1+x^4}\, dx \right]^{-1}$$

$Time's\ up!$

- Answer : 8
- Crucial Step :

Substituting $u = x^2$

Knowing that the antiderivative of $\frac{1}{1+x^2}$ is $\tan^{-1}$.

Using the property $\tan(x) \times \tan\left(\frac{\pi}{2} - x\right) = 1$

A	B	C	D	E	F	G	H	I	J	K	L

Problem 4

04:00

6 pts – Fastest 3, 4 pts – right answerers

$$\left[\frac{1}{\pi}\int_0^{\frac{\pi}{2}} \frac{16}{(1+\cos^2 x)^2}\,dx\right]^2$$

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$$\left[\frac{1}{\pi}\int_0^{\frac{\pi}{2}} \frac{16}{(1+\cos^2 x)^2}\,dx\right]^2$$

$Time's\ up!$

- Answer : 18
- Crucial Step :

  Applying the half-angle formula for the denominator

A	B	C	D	E	F	G	H	I	J	K	L

Problem 5

05:00

8 pts – Fastest 1, 6 pts – right answerers

$$\frac{120}{\pi}\int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}}\frac{1}{(1+x^2)(1+x^7)}dx$$

$$\frac{120}{\pi}\int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}}\frac{1}{(1+x^2)(1+x^7)}dx$$

$$\frac{120}{\pi}\int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} \frac{1}{(1+x^2)(1+x^7)}\,dx$$

$$\frac{120}{\pi}\int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} \frac{1}{(1+x^2)(1+x^7)}\,dx$$

$$\frac{120}{\pi}\int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}}\frac{1}{(1+x^2)(1+x^7)}dx$$

$Time's\ up!$

- Answer : 10
- Crucial Step :

Substituting $y = \frac{1}{x}$ gives

$$2I = \int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} \frac{1}{(1+x^2)(1+x^7)}\,dx + \int_{\sqrt{3}}^{\frac{1}{\sqrt{3}}} \frac{y^2 y^7}{(y^2+1)(y^7+1)} \cdot (-y^{-2})\,dy = \int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} \frac{1}{1+x^2}\,dx\,.$$

A	B	C	D	E	F	G	H	I	J	K	L

Problem 6

03:00

5 pts – Fastest 3, 2 pts – right answerers

$$\int_{-\pi}^{\pi} \frac{\sin(x)\sin(3x)\sin(5x)}{e^{x}+e^{-x}}\,dx$$

$$\int_{-\pi}^{\pi} \frac{\sin(x)\sin(3x)\sin(5x)}{e^{x}+e^{-x}}\,dx$$

$$\int_{-\pi}^{\pi} \frac{\sin(x)\sin(3x)\sin(5x)}{e^{x}+e^{-x}}\,dx$$

$$\int_{-\pi}^{\pi} \frac{\sin(x)\sin(3x)\sin(5x)}{e^{x}+e^{-x}}\,dx$$

$$\int_{-\pi}^{\pi} \frac{\sin(x)\sin(3x)\sin(5x)}{e^{x}+e^{-x}}\,dx$$

$Time's\ up!$

- Answer : 0
- Crucial Step :

  The integrand is an odd function

A	B	C	D	E	F	G	H	I	J	K	L

Problem 7

04:00

6 pts – Fastest 3, 4 pts – right answerers

$$\int_{1}^{e^{\pi}} 2\sin(\log(x)) - 1\, dx$$

$$\int_{1}^{e^{\pi}} 2\sin(\log(x)) - 1\, dx$$

$$\int_{1}^{e^{\pi}} 2\sin(\log(x)) - 1\, dx$$

$$\int_{1}^{e^{\pi}} 2\sin(\log(x)) - 1\, dx$$

$$\int_{1}^{e^{\pi}} 2\sin(\log(x)) - 1\, dx$$

$Time's\ up!$

- Answer : 2
- Crucial Step :

Substituting $u = logx$

Applying integration by parts twice

A	B	C	D	E	F	G	H	I	J	K	L

Problem 8

03:00

$$-1 \leq a, b, c, \ldots, z \leq 1$$

$$Find\ the\ maximum\ of$$

$$\int_{-1}^{1} (x-a)(x-b)\cdots(x-z)dx$$

$$-1 \leq a, b, c, \ldots, z \leq 1$$

Find the maximum of

$$\int_{-1}^{1} (x-a)(x-b)\cdots(x-z)\,dx$$

$$-1 \leq a, b, c, \ldots, z \leq 1$$

Find the maximum of

$$\int_{-1}^{1} (x-a)(x-b)\cdots(x-z)\,dx$$

$$-1 \leq a, b, c, \ldots, z \leq 1$$

$$Find\ the\ maximum\ of$$

$$\int_{-1}^{1} (x-a)(x-b)\cdots(x-z)dx$$

$$-1 \leq a, b, c, \ldots, z \leq 1$$

$$Find\ the\ maximum\ of$$

$$\int_{-1}^{1} (x-a)(x-b)\cdots(x-z)dx$$

$Time's\ up!$

Difficulty : Trivial

- Answer : 0
- Crucial Step :

The integrand =

$$(x-a)(x-b)\cdots(x-x)\cdots(x-z)=0$$

5 pts – Right answer before 30 secs, 3 pts – Fastest 1, 2 pts – Fastest 3, 1 pt – right answerers

A	B	C	D	E	F	G	H	I	J	K	L

Problem 9

04:00

9 pts – Fastest 1, 7 pts – right answerers

$$\left[\int_0^{\frac{\pi}{2}} \sin(314x)\sin^{312} x\, dx\right]^{-1}$$

$$\left[\int_0^{\frac{\pi}{2}} \sin(314x)\sin^{312} x\, dx\right]^{-1}$$

$$\left[\int_0^{\frac{\pi}{2}} \sin(314x)\sin^{312} x\, dx\right]^{-1}$$

$$\left[\int_0^{\frac{\pi}{2}} \sin(314x)\sin^{312} x\, dx\right]^{-1}$$

$$\left[\int_0^{\frac{\pi}{2}} \sin(314x)\sin^{312} x\, dx\right]^{-1}$$

$Time's\ up!$

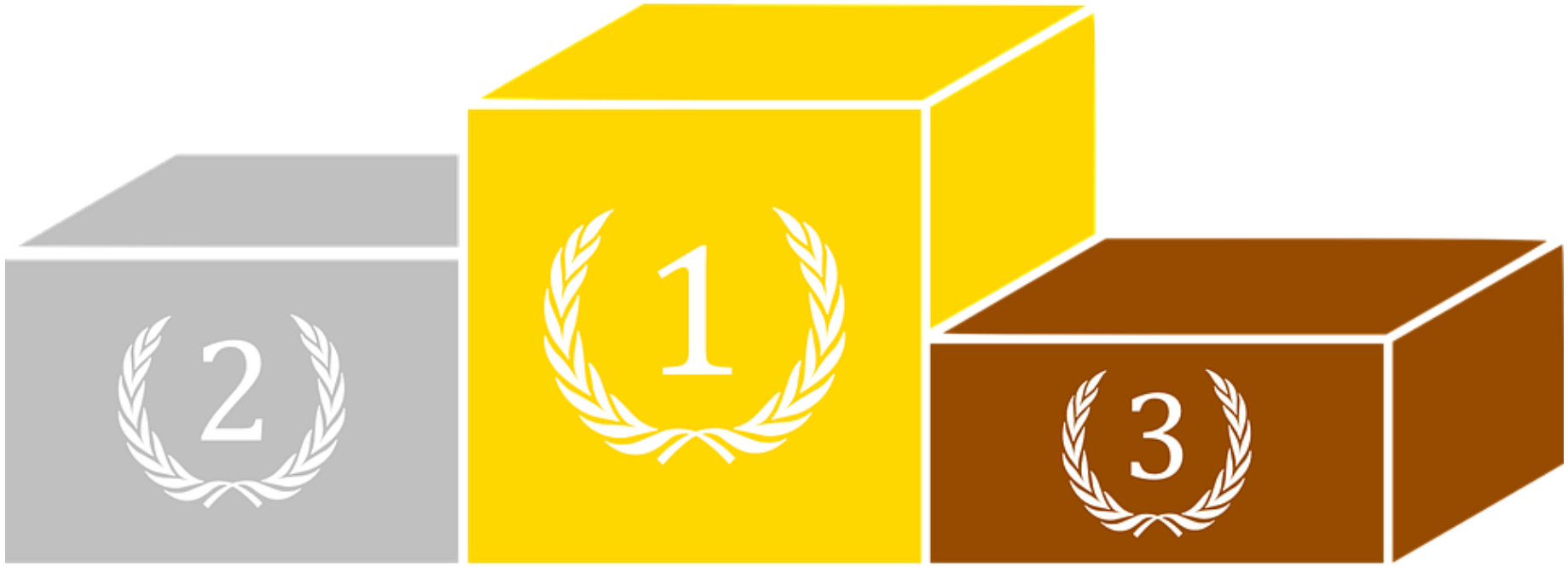
- Answer : 313
- Crucial Step :

$$\sin(314x)\sin^{312}x = \sin(313x)\sin^{312}x\cos x + \cos(313x)\sin^{313}x\,.$$

The antiderivative of the right hand side is $\frac{1}{313}\sin^{313}x\sin(313x) + C$.

A	B	C	D	E	F	G	H	I	J	K	L

Final Result



Congratulations!

Finale